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The oxidation-resistance mechanism of WTaNbTiAl refractory high entropy alloy

Yonggang Yan, Kade A. McGarrity, Daniel J. Delia, Curtis Fekety, Kun Wang *

Kazuo Inamori School of Engineering, Alfred University, Alfred, NY 14802, USA

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ABSTRACT

The WTaNbTiAl alloy demonstrated excellent oxidation resistance with a 31.83 mg/cm^2 of specific mass gain after 48 h oxidation at 1000 °C in the air environment. Based on the multi-scale microstructural characterizations, a diffusion-controlled three-layers oxide scale model was proposed to explain the oxidation kinetics of this RHEA. In addition, thermodynamic calculation results shed light on the formation mechanism of the oxides based on the standard free energy of formation. The present work uncovered the oxidation-resistance mechanism of WTaNbTiAl RHEA via delicate microstructural analysis of oxides, which assists to guide the design of oxidation-resistant high entropy alloys.

1. Introduction

To further improve the efficiency of gas turbines, the application temperatures need to be pushed to higher temperatures, i.e. 1200 ºC or above. However, the current structural materials used in the gas turbine are the limiting factors in raising operation temperatures. Ni-based superalloys, as the current structural component candidate, are limited to operating temperatures of ~1150 °C due to their inherent melting points [1–7]. Therefore, it is critical to explore new high temperature structural alloys superior to current Ni-based superalloys.

Senkov et al [8]. first proposed refractory high entropy alloys (RHEAs) with the inspiration of the idea of high entropy alloys (HEAs), as a potential replacement for the existing superalloys, for application in aerospace, next generation nuclear reactors, and automotive industries etc [9]. One of the main drawbacks of refractory metals (RMs) based alloys is their poor high temperature oxidation resistance a due to the high oxygen activities, which prevents their practical applications as high temperature structural materials. Group IV (Ti, Zr, and Hf) metal elements, have the highest thermodynamic activity towards oxygen at elevated temperatures, yielding extremely poor oxidation resistance performance [10–12]. Regarding the group V metal elements, Nb and Ta would suffer from severe oxidation at temperatures above 400 °C [13], V would be oxidized to be volatile V_2O_5 above 678 °C [14]. It is widely reported that some RMs, such as Ti, Zr, and Hf, could dissolve large amounts of oxygen, which can be present either as interstitially dissolved oxygen atoms or in the form of metal oxides. Generally, these oxides could lead to internal oxidation followed by formation of other complex oxide structures. Additionally, the significant solubility of oxygen in the RMs or alloys would lead to extensive local micro-crack initiation in the oxygen-rich region, which finally leads to partial exfoliation of outer metallic substrate layers. In other words, RHEAs, which have multiple refractory metal elements, would readily form porous and fast-growing oxides or volatile oxides in an oxygen environment especially at high temperatures. This may result in severe degradation of mechanical properties in high temperature and oxygen environments. Despite numerous efforts which have been made on the RHEAs research [15–23], the application of RHEAs at elevated temperatures is still limited mainly due to their poor high temperature oxidation resistance, which becomes worse with increasing temperature. Therefore, the development of new high temperature oxidation-resistant RHEAs and understanding the underlying oxidation mechanisms are critical.

To the best of our knowledge, the oxidation resistant mechanism of the current RHEAs is the formation of complex dense protective oxide scales on the surface of the materials to retard further oxidation of the metallic substrate. Thus far, the addition of non-refractory metals such as Cr and Al, or Si, seems to be the most promising method to improve high temperature oxidation resistance of RHEAs [24–28]. For instance, Gorr et al [29]. reported that the addition of Si into NbMoAlCrTi RHEA may improve its oxidation resistance. Si can form a protective SiO_2 scale at the interface between the metallic substrate and oxide, leading to the reduced inward diffusion of oxygen. Further, SiO_2 acts as nucleation sites for the formation of a homogenous protective oxide scale, which

* Corresponding author.
E-mail address: wang@alfred.edu (K. Wang).

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inhibits the inward and outward diffusions of oxygen and metal cations, respectively, which aids in achieving better oxidation resistance. Senkov et al [30]. studied the oxidation behavior of $NbCrMo_{0.5}Ta_{0.5}TiZr$ at 1000 °C in air. The specific mass gain of the alloy is about 120 mg/cm^2 after 100 h of oxidation at 1000 °C, showing excellent oxidation resistance over conventional Nb-based alloys. Lo et al [31]. reported $CrTaO_4$-based oxide scale would form on the surface of AlSiTiCrNbMoTa RHEAs system after oxidation at 1000 °C and 1100 °C in air for 200 h. Gorr et al [32]. also examined that the formation of $CrTaO_4$ external scale could reliably protect the TaMoCrTiAl from being oxidized in a temperature range of 500–1500 °C. For the current reported oxidation resistant alloys, there is very little work based on nanoscale analysis to reveal the oxidation resistant mechanisms of RHEA. In addition, these oxidation resistant alloys usually formed intermetallic secondary phases [33–35], which tend to sacrifice the mechanical properties of the alloys. Thus, the design of single-phase oxidation resistant RHEAs and study of its oxidation resistant mechanism would speed up the practical application of oxidation resistant alloys.

In this work, W, Ta, Nb, Al, and Ti were selected to design high temperature oxidation resistant alloy. W, Nb and Ta were chosen due to their high melting points, while the addition of Al and Ti enables the formation of protective oxide scales and reduces the density of alloy. Our previous machine learning results [36] have predicted the favorable formation of novel equiatomic single-phase RHEA WTaNbTiAl. The oxidation resistant mechanism of as-prepared equimolar RHEA was interrogated based on the nanoscale microstructural analysis.

2. Experimental

In this study, high purity raw W (99.9%, 1–5 μm), Ta (99.8%, 1–5 μm), Nb (99.8%, 1–5 μm), (purchased from Atlantic Equipment Engineers), Al (99.5%, −325 mesh), and Ti (99.5%, −325 mesh) powders (purchased from Fisher Scientific) were used. The equimolar metallic powders were mixed via planetary ball milling for 2 h with the milling speed of 200 revolutions per minute (RPM). The as-milled RHEAs powders were characterized using an X-ray Diffractometer (XRD, BRUKER D2-ADVANCE) with CuKα radiation. Next, the as-mixed powders were sintered via SPS (HPD 25, FCT Systeme GmbH, Germany) at 1400 °C for 20 min under a uniaxial pressure of 98 MPa in vacuum. As-prepared samples were polished to 2000 grit to observe their microstructures.

The immersion method was used to measure apparent bulk densities for all samples in deionized water. Shear modulus and elastic modulus (E) were measured via the pulse echo technique, which operates on the principle of elastic wave propagation and reflection in a solid. Vickers hardness test was performed with a (V-100 A2 Hardness Tester, Leco Corporation, St. Joseph, MI). Measurements were made under a 1 kgf load for 15 s. Five indentations were used to calculate an average Vickers hardness value in advisement with ASTM E92–17. Finally, oxidation studies were conducted with thermogravimetric analysis (TGA) at 1000 °C in air. Isothermal oxidation tests were also conducted at 1000 °C for different durations (0.5 h, 12 h, 24 h, 36 h and 48 h) in a muffle furnace with a heating and cooling rate of 200 °C/h. The samples were analyzed using scanning electron microscopy (SEM), X-ray diffraction (XRD) before and after oxidation experiment.

Thermo Fisher Scientific (FEI) Scios™ 2 DualBeam ultra-high-resolution analytical FIB-SEM system was used to prepare the transmission electron microscopy (TEM) samples. (Scanning) transmission electron microscope ((S)TEM) and the EDX elemental analysis were conducted using Thermo Fisher Scientific (FEI) Talos F200X equipped with 4 in-column SDD Super-X detectors. The standard free energy of formation of the oxides in the range between 100 °C and 1000 °C were calculated using the commercial CALPHAD (CALculation of PHAse Diagrams) software FactSage V8.0 (Thermfact/CRCT, Montreal, Canada and GTT-Technologies, Aachen, Germany).

3. Results

3.1. As-prepared alloy

Planetary ball milling was performed with 200 rpm (revolution per minute) milling speed for two hours to avoid introducing significant contaminations. Therefore, the powders were not completely alloyed after ball milling. After SPS sintering, the material was fully alloyed and formed a single BCC phase. XRD patterns of the mixed metal powders and sintered RHEAs can be found in Fig. S1 in the supplemental document. Peak broadening was observed in the XRD pattern of the as-prepared RHEA WTaNbTiAl. The broadening is believed to be caused by the lattice distortion due to the formation of the single-phase high entropy alloy after the SPS sintering. The SEM EDX element mapping of the as-prepared RHEA WTaNbTiAl is displayed in Fig. S2 and S3. The microstructure uniformity is fairly good, although the minor phases also exist in localized regions. The minor phases include W-rich and Ti-rich segregation phases which form during the sintering process, as well as the dispersed alumina and titanium oxide phases which are due to the oxygen introduction during balling milling process [37]. Thus, WTaNbTiAl alloy has a disordered, ideally single-phase BCC, solid solution microstructure. Although the dispersed alumina and titanium oxide phases may have an effect on the subsequent oxidation experiment, the focus of this work is on the oxidation behavior of the primary phase. The apparent bulk density of the sample is 10.21 g/cm^3. No pores were observed from the SEM observations while apparent porosity was determined via immersion method to be 0.05%, qualitatively indicating that the RHEA was fully densified during the sintering. WTaNbTiAl alloy has a lattice constant value of 3.23 Å calculated by JADE. The elastic moduli and shear moduli of the RHEA are 171 GPa and 65 GPa, respectively. The mean Vickers hardness value of the as-prepared RHEA is 6.7 ± 0.9 GPa. The average grain size of the as-prepared WTaNbTiAl sample is ~ 29 μm after statistical measurement.

3.2. Oxidation behavior at 1000 °C

Oxidation behavior is one of the crucial technical metrics for high temperature structural materials. According to the image of the WTaNbTiAl RHEA after 12 h of oxidation, as shown in Fig. 1, WTaNbTiAl alloy is capable of withstanding oxidation at 1000 °C. Continuous mass gain occurred during holding WTaNbTiAl alloy at 1000 °C in air. The mass gain and heat flow versus time curves were obtained via TGA measurement, as displayed in Fig. 1. The mass gain followed a two-step process, which can be fitted with the following equation,

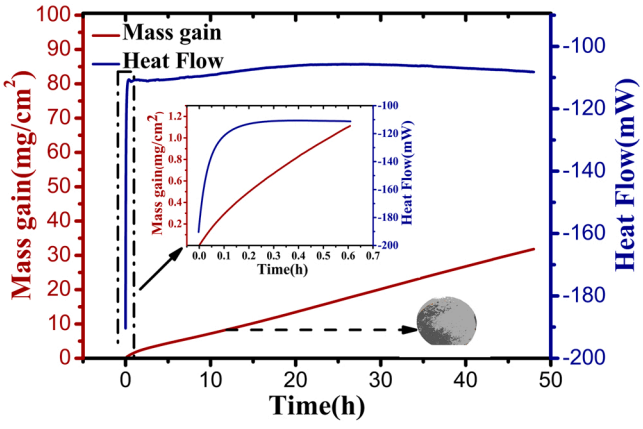


Fig. 1. Mass gain and heat flow curves during exposure to air at 1000 °C. The insert photo is the physical shape of the WTaNbTiAl alloy after oxidation at 1000 °C with heating rate of 200 °C /h to the oxidation temperature for 12 h.

$$\frac{\Delta m}{A} = \begin{cases} \ln(0.35t) & 0 < t < 0.5 \\ 0.65t + 0.81 & 0.5 < t < 48 \end{cases} \quad (1)$$

Where $\frac{\Delta m}{A}$ represents mass change per unit area, Δm is mass gain, A is surface area of the sample, t is exposure time (h).

The trend of mass gain curve in current work did not obey the existing oxidation kinetics law in previous studies, such as exponential kinetics [31,38,39], parabolic kinetics [40], etc., as shown in Fig. 1. In the early oxidation stage (0 < t < 0.5h), WTaNbTiAl RHEA showed mass gain with a logarithmic rate, which then decreased to a near constant oxidation rate (0.5h < t < 48h), yielding a very low mass gain (31.83 mg/cm^2) up to 48 h. The negative heat flow values indicate the exothermic reaction. The absolute value of heat flow decreased dramatically in the first several minutes, and then decreased slowly, slightly increased after approximately 20 h of exposure, displaying the decrease of oxidation rate occurred when the oxidation time extended up to 20 h. Generally, the moderate mass gains indicated slow oxidation rate of the WTaNbTiAl alloy after 48 h of exposure to air at 1000 °C. The oxidation-resistance performance of WTaNbTiAl in this work is also compared with other reported RHEAs with exceptional oxidation-resistance, which can be seen in Fig. 2.

Mirror-polished cross-section surfaces of WTaNbTiAl alloy after oxidation at 1000 °C with different dwelling times are depicted in SEM backscattered electron (BSE) images in Fig. 2. Interestingly, the oxide layers at all oxidization conditions tends to display a similar three-layers structure, which will defined as outer oxide layer (OOL), an internal oxidation front (IOF), and an internal "partially oxidized" layer (POL). After 12 h of exposure to air at 1000 °C, a ~120 μm thick oxide layer that consists of an approximately 10 μm OOL, 90 μm IOF layer, and 20 μm POL, covered the surface of the RHEA WTaNbTiAl, as shown in Fig. 3(a). The POL in RHEAs was caused by the high oxygen solubility and reactivity between the RMs and oxygen at elevated temperatures [49]. As the oxidation time increased from 12 h to 48 h, the thickness of the oxide layer increased from 120 μm to 400 μm, as shown in Fig. 3(b) and (c). In general, the oxide layer tends to serve as a reliable diffusion barrier and effectively isolates the substrate from oxygen [24]. Thus, the oxidation rate of the metallic substrate is decreased. The oxide layer of the RHEA after 24 h of oxidation would exfoliate from the metallic substrate during the cooling process from oxidation temperature to room temperature, which is presumably due to different coefficients of thermal expansion and specific volume of the oxide and the alloy [30].

XRD patterns of WTaNbTiAl alloy (Fig. 4) from mixed powder to 48 h of exposure to air at 1000 °C demonstrates phase evolutions on the surface of the WTaNbTiAl alloy during the oxidation process. The formation of oxide polymorphs leads to anisotropic expansion, thus resulting in the lattice strain of the oxides. This results in slight peak shift and peak broadening of the primary XRD peaks in the first 36 h oxidation hours. For the oxidized WTaNbTiAl alloy, the peak intensity of the pattern for 0.5 h of oxidation is different from that after 12 h of oxidation, while the peaks almost distribute at the same positions. In addition, the peaks intensity of the XRD patterns for those dwelling over 12 h are almost the same. The species of the produced oxides would not change after 0.5 h of oxidation, while their proportions would continuously change in the first a few dwelling hours, and finally approach a constant after 12 h. XRD is limited by the penetration depth therefore only the oxide layer phase is detected. To identify the phase of the entire oxide layer, XRD measurements as shown in Fig. 5 and EDX elemental mapping (see Fig. S4) were performed on internal and inner side of the spalled-off oxide layer (36 h oxidation), respectively. In addition, Fig. 6 shows the cross-sectional BSE images and the EDX elemental mapping of WTaNbTiAl alloy after 48 h of oxidation. The XRD and SEM analyses reveal that the main phases of the OOL are $AlNbO_4$, Ti-rich oxides based on the rutile structure (eg. $Ti_{0.33}Al_{0.33}Ta_{0.33}O_2$) and $Ta_{1.5}Nb_{1.5}O_3$. It is believed that this oxide layer could serve as an effective barrier to the inward diffusion of oxygen, thus retarding the oxidation of RHEA WTaNbTiAl. Regarding the IOF, it mainly consists of $Ta_8W_9O_{47}$ and some Ti-rich oxides, as shown in Fig. 5. Phases at the bottom of IOF (inner side of the spalled-off oxide layer) are more complex. The composite oxides mainly consist of $3WO_3{\bullet}Ta_2O_5$, $15Ta_2O_5{\bullet}2WO_3$, $TiTa_2O_7$, $TaO_{1.7}$, Ta_2O_5, WO_3, $AlTaO_4$ and $(AlTiTa)_{0.677}O_4$. The quantified chemical compositions (in at%) based on STEM-EDX mapping also support that the oxide layer has a thin OOL ($AlNbO_4$ and Ti-rich oxides) and IOF ($Ta_8W_9O_{47}$ and Ti-rich oxides), as shown in Figs. 7–9.

To further understand the oxidation mechanisms in nanoscale, TEM analysis was conducted on samples extracted from different depth sites in the oxide layer after 36 h of oxidation at 1000 °C. According to XRD patterns and TEM analysis of the oxide layer, as shown in Figs. 4 and 5 and Figs. 7–9, it was observed that the oxides species and the proportions of the oxides tended to vary significantly at different depths of the oxide layer. For OOL, STEM-HAADF (high angle annular dark field) image and the corresponding STEM-EDX elemental mapping show the OOL mainly consists of AlNb-rich oxide phase, Ti-rich oxide phase, and TaNb-rich oxide phase (Fig. 7). These results combined with XRD analysis, confirmed that these phases are $AlNbO_4$ (area 1 and 3), Ti-rich oxides (area 4), and $Ta_{1.5}Nb_{1.5}O_3$ (area 2). The quantitative analysis results are also shown in Fig. 7, suggesting that the amount of the main phases obeys the following order: $AlNbO_4$ > Ti-rich oxides > $Ta_{1.5}Nb_{1.5}O_3$. The STEM-HAADF image presented in Fig. 8, the corresponding STEM-EDX elemental mapping results, and quantitative analysis results of the internal oxide front (IOF) layer. It is shown that the IOF is composed of WTaNb rich oxide phase and Ti-rich oxide phase. In addition, Al homogeneously distributes in the oxide matrix. By combination of the quantitative analysis results and XRD investigations (Fig. 5), it was found that only $Ta_8W_9O_{47}$ (area 1 and 3) and Ti-rich oxides (area 2) were observed in IOF, no complex oxides (composed of tungsten, niobium, and tantalum) or Al-containing oxides were observed. Fig. 9 displayed the TEM investigations of the inner side of the exfoliation oxide layer. This region is mainly composed of WTaNb-rich and AlTi-rich areas. The combination of XRD pattern (Fig. 5) and quantitative analysis results suggests that the main phases in this position are very complex, mainly consists of tungsten tantalum oxide and AlTi-rich oxides phase. The grain size at this position (average diameter of 140 nm) is much smaller than that in the IOF layer (average diameter of 571 nm) and the OOL layer (average diameter 380 nm), which means grain grew to larger size with oxidation time.

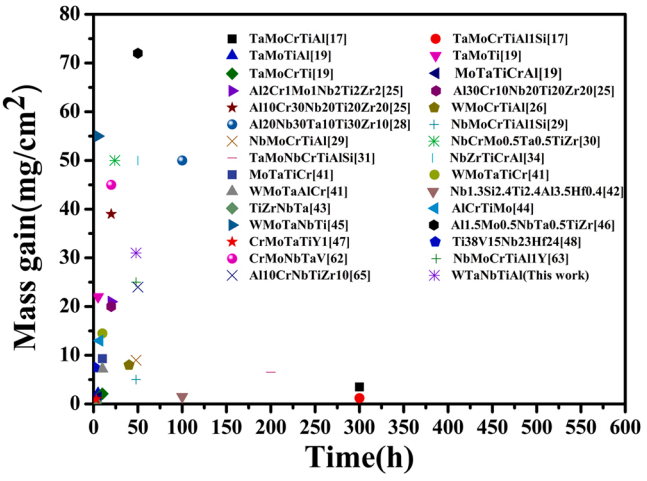


Fig. 2. A comparison of mass gain between WTaNbTiAl in this work and other RHEAs reported previously [41–48].

4. Discussion

4.1. Oxidation resistant mechanism

The oxidation behavior of RHEAs depends on the phases and

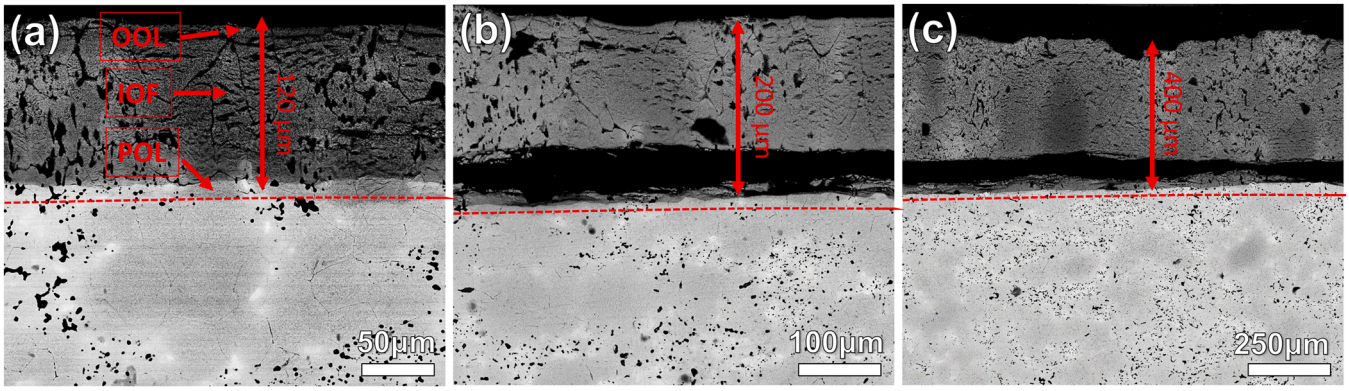


Fig. 3. Cross-sectional BSE images of the RHEA WTaNbTiAl after oxidation at 1000 °C for 12 h (a), 24 h (b) and 48 h (c), respectively.

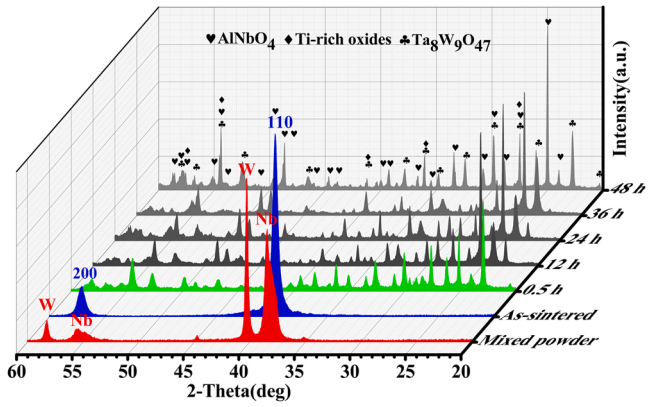


Fig. 4. XRD patterns of WTaNbTiAl alloy from mixed powder to 48 h oxidation, respectively.

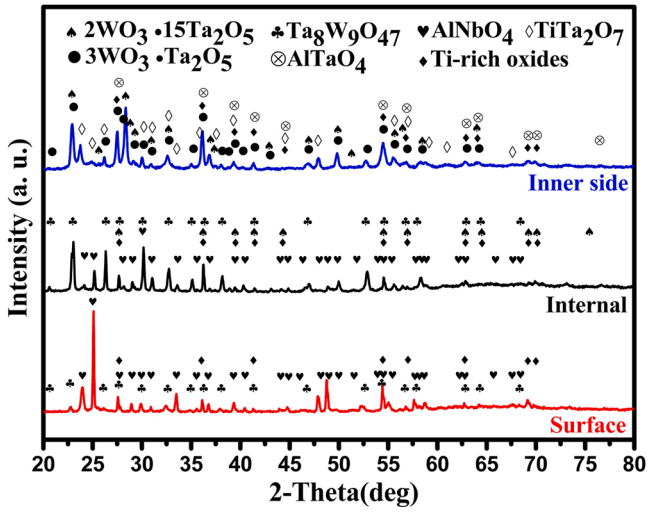


Fig. 5. XRD patterns of the spalled-off oxide layer at different positions after 36 h of oxidation.

microstructures of oxide layer [24,50]. In this work, the unique oxidation kinetics of WTaNbTiAl alloy was assumed to be caused by its initial microstructure [28,51] and chemical composition. The latter determines the phase formation of the oxide layer, which directly affects the oxidation behavior of the RHEA. The SEM observations Fig. 10 (b) of the micron-scale pores in the oxide layer, which could serve as fast migration channels for oxygen and cations. In addition, these pores would develop to cracks, and lead to partial exfoliation of oxide layer. Accordingly, these pores would affect the oxidation rate of the metallic substrate, contributing to the unique mass gain curve of the RHEA WTaNbTiAl. Also, it's worthwhile to mention that a higher-purity WTaNbTiAl sample is anticipated to have superior oxidation-resistance as compared to the sample in the present work. In combination with BSE images and EDX elemental mapping results, the pores were believed to be caused by impurities introduced during material fabrication process, which are mainly alumina, titanium oxide, as shown in Figs. S2, S3 and Fig. 10(a). Generally, the raw metal powders would absorb oxygen when exposed to air. In addition, oxygen could be dissolved into the metal powder during planetary milling. The dissolved oxygen would react preferentially with Ti and Al to form oxides due to their more negative standard free energy of formation, and finally grow as Al_2O_3 and TiO_2-rich zones in the alloy substrate.

Based on the identified microstructures and phases in the oxide layer, the oxidation process of WTaNbTiAl can be modeled as diffusion-controlled growth of multilayer oxide scale, as shown in the schematic of Fig. 11. The RHEA WTaNbTiAl was found to form a variety of complex refractory oxides (e.g. $AlNbO_4$, $Ta_8W_9O_{47}$), along with simple oxides (e.g. $TaO_{0.16}$, WO_3 and TiO_2). The oxides species and their proportions depend on their positions in the oxide layer, as shown in the Figs. 7–9. From the XRD and TEM investigations of the inner surface of the IOF, the formed oxides are complex, which are mainly tungsten tantalum oxides ($3WO_3$•Ta_2O_5, $Ta_8W_9O_{47}$) and titanium aluminum oxides. Some simple oxides, like Ta_2O_5 and WO_3, were also observed. The oxide grain size in IOF increased from nanometer level (average diameter 140 nm) at its IOF-POL region to the sub-micrometer (average diameter 0.57 μm) at its region. This also supports the conclusion that the phase evolution continuously occurred in this layer. From the XRD pattern of IOF, the crystal structure of the oxide in IOF is $Ta_8W_9O_{47}$ as well, which has a tetragonal tungsten bronze type phase (TTB) structure. Actually, this oxide is $Nb_xTa_8W_9O_{47}$ instead of $Ta_8W_9O_{47}$ in area 1 of Figs. 8 and 9. Generally, TTB consists of a framework of MO_6 corner-sharing octahedra linked in such a way that three-, four- and five-sided tunnels are formed [52]. Fig. 12(a) is the high-resolution STEM image of the $Nb_xTa_8W_9O_{47}$, while Fig. 12(b) is the schematic of this unique TTB oxide structure. Nb cation as well as other cations seems to insert in these tunnels, but the crystal structure is unchanged [53], as shown in Fig. 12 (b). This explains why the Nb atoms homogeneously distribute in TaW-rich regions according to the STEM-EDX elemental mapping results in Fig. 8. These tunnels serve as paths for both inward diffusion of the oxygen atoms and outward diffusion of the Al, Nb and Ti cations, and finally forming the protective outer layer ($AlNbO_4$ and Ti-rich oxides) on the surface of the oxide layer (OOL). Different from TTB structure of $Nb_xTa_8W_9O_{47}$, $AlNbO_4$ has more condensed monoclinic crystal structure as shown in the high-resolution STEM images in Fig. S5 in the supplemental materials, which serves as a more effective protection layer for oxygen inward diffusion.

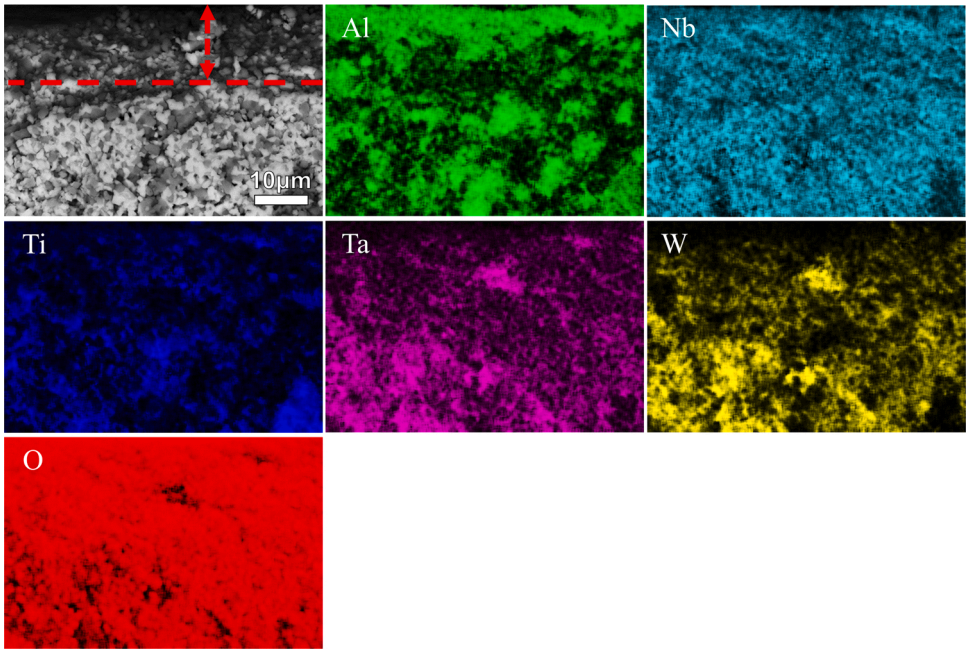
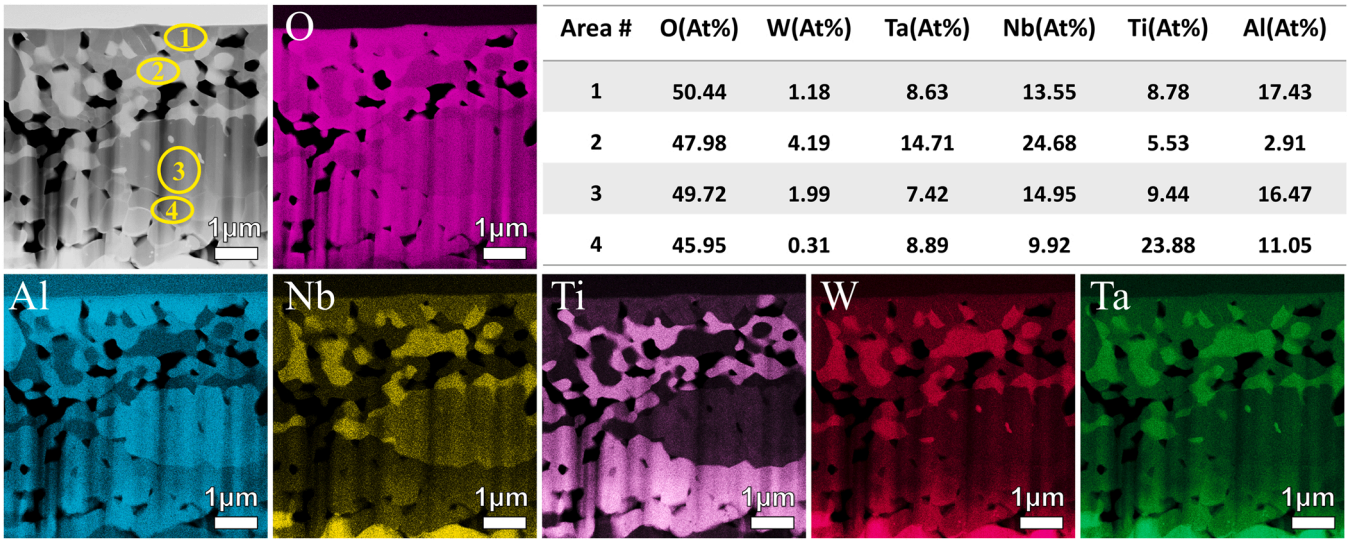


Fig. 6. The cross section of oxide layer after 48 h oxidation. The red dash line indicates the outer composite oxide layer (OOL).



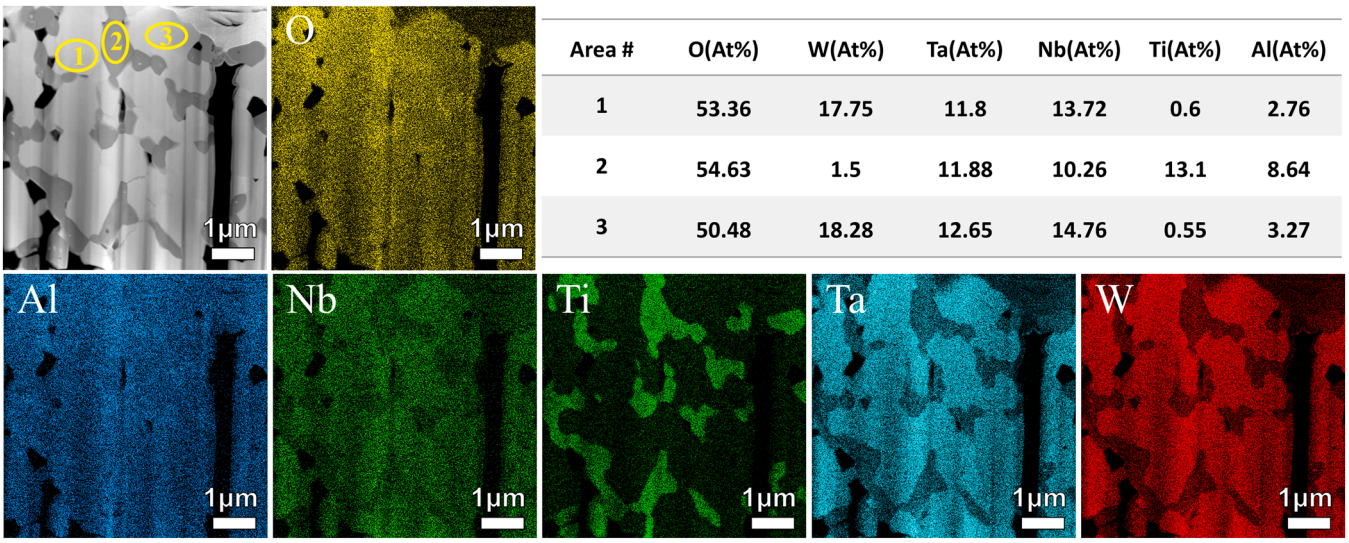
Area #	O(At%)	W(At%)	Ta(At%)	Nb(At%)	Ti(At%)	Al(At%)
1	50.44	1.18	8.63	13.55	8.78	17.43
2	47.98	4.19	14.71	24.68	5.53	2.91
3	49.72	1.99	7.42	14.95	9.44	16.47
4	45.95	0.31	8.89	9.92	23.88	11.05

Fig. 7. Bright field (BF) TEM image of oxide layer surface after 36 h of oxidation, the corresponding STEM-EDX elemental mapping results of the BF TEM image, and the quantitative analysis results for the selected areas.

The decrease of Al, Nb and Ti through the oxide layer from outer surface to inner surface also supports the diffusion-controlled growth model. Regarding the concentration of each element, the amounts of Al, Nb and Ti in the IOF are relatively lower than those in OOL, while the W and Ta contents in the IOF are relatively higher than that in the OOL, as shown in table in Fig. 8. This also supports the assumption that the Al, Nb and Ti cations diffuse to the surface of the oxide layer and form $AlNbO_4$ and Ti-rich oxides based OOL.

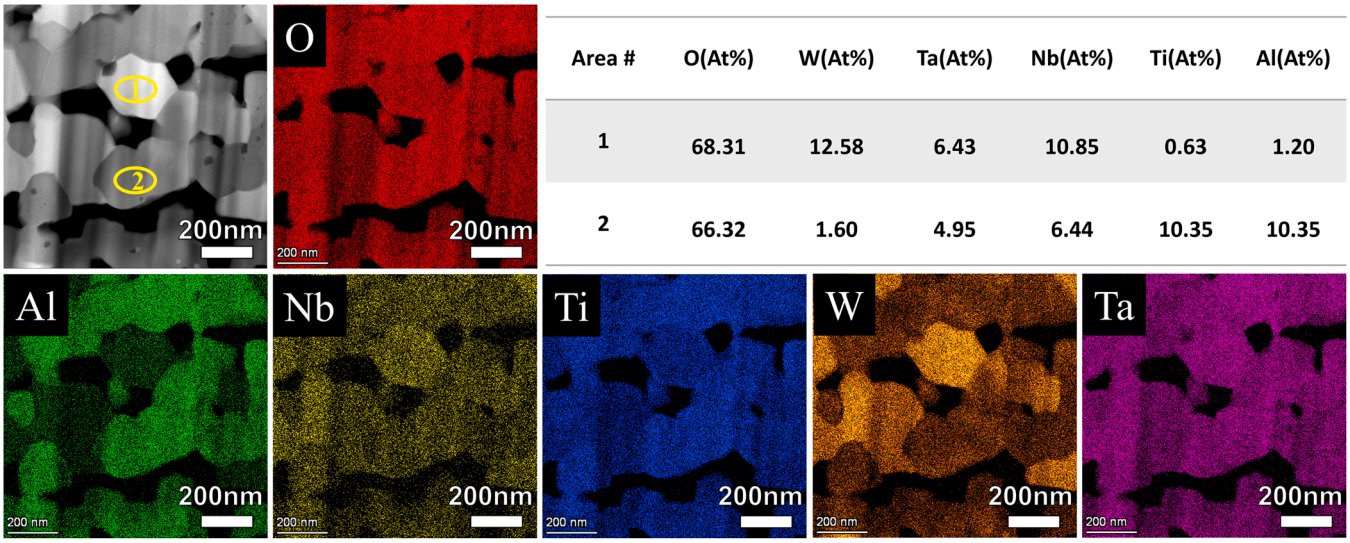
The thickness of each layer reveals that the OOL is potentially the primary inhibiting factor of the inward oxygen flux and the corresponding oxidation kinetics. Therefore, the mass gain dramatically increases at the early oxidation stage, as shown in Fig. 1. After holding the oxidation temperature for longer than 30 mins, the protective OOL formed, which is consistent with the XRD investigations of the oxide layer surface, as shown in Fig. 4. The formed OOL impedes the inward diffusion of oxygen and slows down the growth rate of the POL, leading to a suppressed oxidation rate of the sample. Thus, the slope of the mass gain curve decreases with the oxidation time and then tend to be constant. Overall, the above results suggest that the OOL of the RHEA WTaNbTiAl facilitates a suppression in the oxidation kinetics and provides moderate mass gain up to 48 h, which subsequently appear to offer resistance to inward diffusion of oxygen and outward diffusion of cations.

Specific to the given alloy system, the unique species and proportion of the oxides that forms in the oxide scale do influence their oxidation kinetics. Butler and Chaput [28] observed the formation of $AlNbO_4$-based oxide layer after a prolonged exposure of $Al_{20}Nb_{30}Ta_{10}Ti_{30}Zr_{10}$ in air at 1000 °C, and pointed out that the $AlNbO_4$-based oxide layer protects the metallic substrate from being further oxidized. Jayaraj et al [54]. reported that the formation of complex oxides of $AlNbO_4$ and Ti_2ZrO_6 on the surface of AlNbTiZr HEA at 1000 °C can lead to the sluggish oxidation kinetics. Allouard et al [33]. investigated the



Area #	O(At%)	W(At%)	Ta(At%)	Nb(At%)	Ti(At%)	Al(At%)
1	53.36	17.75	11.8	13.72	0.6	2.76
2	54.63	1.5	11.88	10.26	13.1	8.64
3	50.48	18.28	12.65	14.76	0.55	3.27

Fig. 8. Bright field (BF) TEM image of internal oxide front (IOF) layer after 36 h of oxidation, the corresponding STEM-EDX elemental mapping results of the BF TEM image, and the quantitative analysis results for the selected areas.



Area #	O(At%)	W(At%)	Ta(At%)	Nb(At%)	Ti(At%)	Al(At%)
1	68.31	12.58	6.43	10.85	0.63	1.20
2	66.32	1.60	4.95	6.44	10.35	10.35

Fig. 9. Bright field (BF) TEM image of inner side of the exfoliated oxide layer after 36 h of oxidation, the corresponding STEM-EDX elemental mapping results of the BF TEM image, and the quantitative analysis results for the selected areas.

influence of Al concentrations on the oxidation resistance of NbTiAl alloy. The authors observed that the protective oxide $AlNbO_4$ would form as the content of Al increases, resulting in a reduction in oxidation rate compared to lower Al containing NbTiAl alloy that mainly formed $TiNb_2O_7$.

Thus far, limited results have been reported on the protectivity of WTa oxides. To confirm which oxide is the main phase contributing to the oxidation resistance of the RHEA WTaNbTiAl, a binary WTa was synthesize by planetary ball milling and SPS consolidation. Likewise, oxidation experiment was performed at 1000 °C for 2 h and 8 h, respectively. Loose oxide layer mainly contains $Ta_8W_9O_{47}$ (see supplementary Figs. S6 and 7) formed after 8 h of oxidation, suggesting that $Ta_8W_9O_{47}$ oxide is ineffective for protecting the alloy substrate. It is believed that $Ta_8W_9O_{47}$ would not inhibit the diffusion of oxygen and cations due to its unique tunnel structure.

4.2. The formation mechanism of the oxides

Regarding the oxide formation mechanisms of the RHEA, it is believed to be controlled by oxygen-chemisorption equilibrium, nucleation and growth of oxides based on the oxygen solubility in the alloy and standard free energy of formation of the oxides. The strong tendency of internal oxidation in RHEAs could be related to the high solubility, diffusivity and reactivity of many of the principal metal elements with oxygen [55]. As mentioned before, the refractory metals are intrinsically prone to dissolving notable amounts of oxygen. The oxygen solubility of each component in RHEA WTaNbTiAl is shown in Table 1, suggesting that RHEA WTaNbTiAl has a relatively high oxygen solubility primarily due to the Ti, Ta and Nb.

Compared to pure metals, RHEA will form complex oxides at elevated temperatures, because each component will form several unstable or stable oxides. Some of the oxides are volatile products, which will sublimate at high temperatures. In addition, the formed simple oxides will continuously react/combine into complex oxides. Ti in Ti-containing RHEA tends to form only one thermodynamically stable solid oxide at 1000 °C in air (TiO_2), instead of complex oxides [17,26,34,54,57–60]. In current work, the TiO_2 directly develops as TiO_2-rich regions, as showed in the Fig. 7. Müller et al [50]. pointed out that Ti

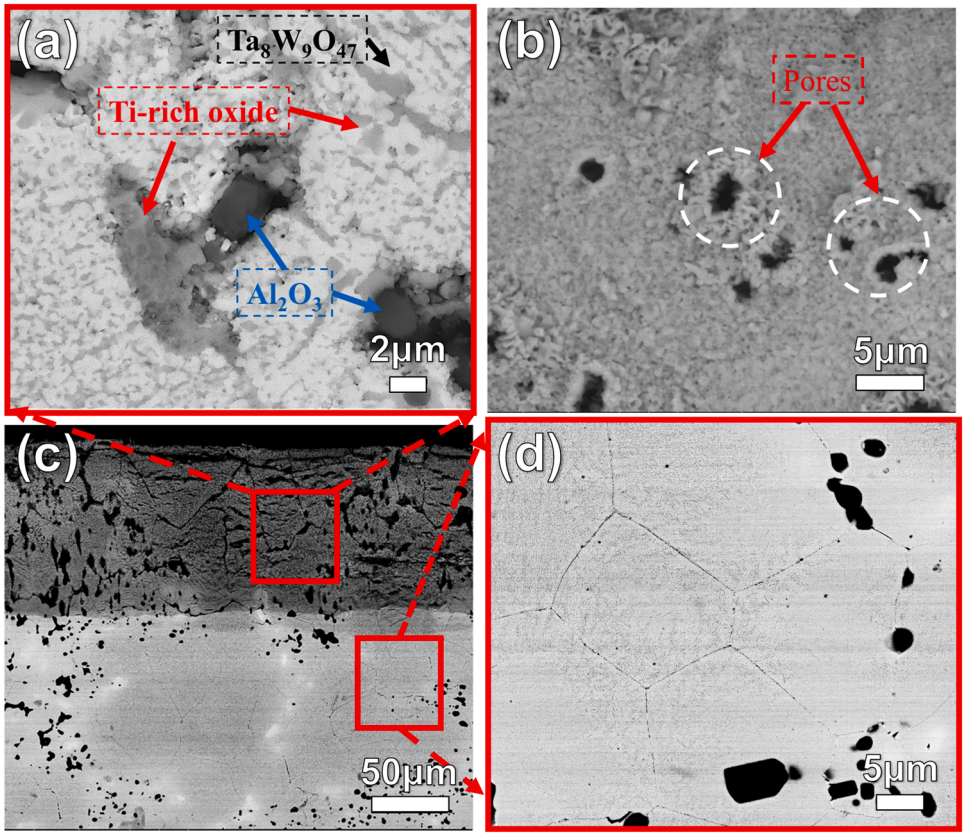


Fig. 10. SEM images of (a) the cross section of the oxide layer and (b) inner surface of the spalled-off oxide layer, (c) the cross section of the oxidized sample, (d) the alloy substrate.

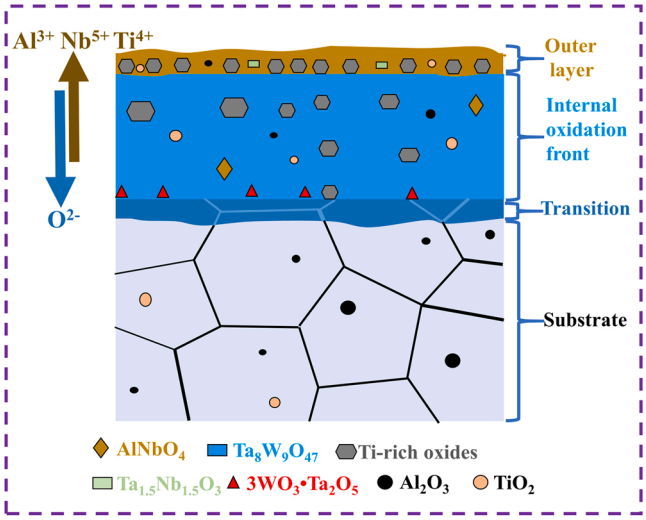


Fig. 11. Schematics of oxidation process of RHEA WTaNbTiAl at 1000 ℃. The OOL is the $AlNbO_4$ and Ti-rich oxides region, which is formed by the diffused Nb, Al and Ti cations after an extensive oxidization time. The IOF is the $Nb_xTa_8W_9O_{47}$ region, which acts as a tunnel for the inward diffusion of oxygen and outward diffusion of the cations. The POL is the innermost region of the oxide layer that substrate is partially oxidized by the diffused oxygen.

seems to improve oxidation resistance of the RHEA according to the comparison of oxidation behavior of Ti-containing RHEAs (NbMoCrAlTi and TaMoCrAlTi) and their Ti-free systems. Regarding Ta in Ta-containing RHEA, only one thermodynamically stable solid oxide, Ta_2O_5, can form at 1000 °C, but Ta_2O_5 tends to react with other oxides to form complex oxides [61]. The oxidation behavior of Nb in Nb-containing RHEA is in many ways analogous to that of Ta, but it is even more complex due to the fact that Nb forms several stable oxides: NbO[54]or NbO_2 [62], which have the rutile structure, as well as several polymorphs of Nb_2O_5 [25,50,63,64], and complex oxides [14,18,25,28,30,34,54,62,65]. The oxidation behavior of W-containing HEAs at 1000 °C is rarely reported. W seems to be oxidized as WO_{3-x} and subsequently form complex oxides [26,61].

During the initial stage, oxidation of the RHEA WTaNbTiAl proceeds principally by the dissolution of oxygen in the alloy, and this is subsequently followed by the formation of simple oxides. As oxidation proceeds, it is believed that simple oxides form complex oxides by a diffusion-controlled process. The standard free energy of formation for oxides have been widely used to assess the formation of favorable oxides in many previous works [29,34,50,54]. In doing so, the standard free energy of formation of relevant oxides in this work in the range between 100 °C and 1000 °C were calculated using the commercial CALPHAD (CALculation of PHAse Diagrams) software FactSage V8.0, as shown in Fig. 13. The calculated ΔG^0 of the simple oxides at 1000 °C are summarized in Table 2.

The simple oxides, like Al_2O_3, TiO_2, Ta_2O_5, Nb_2O_5, and WO_3, will form at the early stage. From the standard free energy of formation values, it can be seen that the formation of favorable oxides follows the order Al_2O_3 >TiO_2 >Ta_2O_5 >Nb_2O_5 >WO_3. The nucleation of Al_2O_3 and TiO_2 happens prior to other RM oxides. This conclusion is consistent with that in [24,34,50].

Despite that RHEA WTaNbTiAl is Al-containing alloy, no dense alumina scale is observed in current work. This is because the formed Al_2O_3 in Al-containing RHEA readily forms complex oxides,like $AlTiO_5$ [65], $AlNbO_4$ [14,60] or $Al_2(WO4)_3$ [26]. Intensive effort has been made to develop RHEAs that can form dense continuous Al_2O_3 layers, but no successful case has been reported thus far.

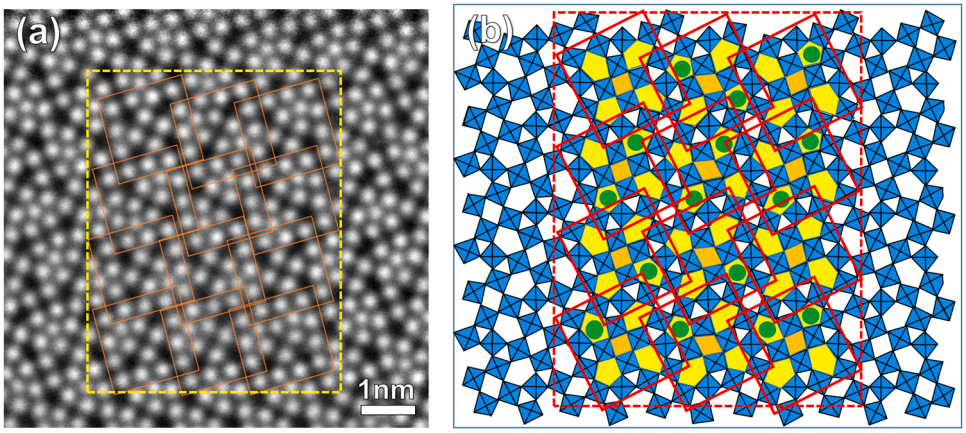


Fig. 12. (a) High-resolution HAADF-STEM of $Nb_xTa_8W_9O_{47}$, (b) The schematic model of $Nb_xTa_8W_9O_{47}$. Unit cells are framed. A cell of the tetragonal tungsten bronze type (TTB; red box) have four pentagons of MO_6 octahedra (M = Ta, W) arranged around a central square (Orange). Some of pentagonal tunnels are systematically occupied by metal-oxygen strings (filled circles).

Table 1
Oxygen solubility in pure elements at 1000 °C relevant for RHEA.

Element	W	Ti	Al	Nb	Ta
O solubility, [at%]	0.03[26]	14[11]	~0[26]	2.5[24]	3[56]

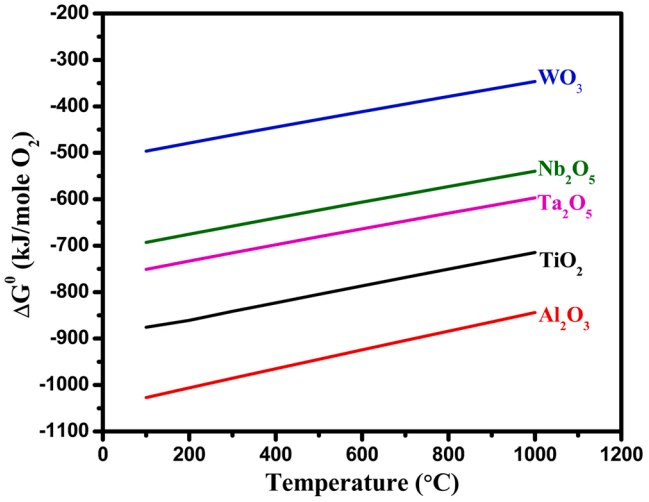


Fig. 13. Standard free energy of formation of relevant simple oxides as a function of temperature.

Table 2
Standard free energy of formation of relevant oxides at 1000 °C.

Oxide	Ta_2O_5	TiO_2 (rutile)	Al_2O_3	Nb_2O_5	WO_3
ΔG^0[kJ/mole O_2]	− 598	− 713	− 853	-540	-346

For RHEA WTaNbTiAl, the main oxides in the whole oxide layer consists of $AlNbO_4$, Ti-rich oxides, and $Ta_8W_9O_{47}$. For these oxides, standard free energies of formation at 1000 °C can be calculated according to following equations [50].

$$\frac{1}{2}Al_2O_3(s) + \frac{1}{2}Nb_2O_5(s) = AlNbO_4(s) \quad (2)$$

$$\frac{1}{2}Al_2O_3(s) + \frac{1}{2}Ta_2O_5(s) = AlTaO_4(s) \quad (3)$$

$$9WO_3(s) + 4Nb_2O_5(s) = Nb_8W_9O_{47} \quad (s) \quad (4)$$

$$9WO_3(s) + 4Ta_2O_5(s) = Ta_8W_9O_{47} \quad (s) \quad (5)$$

$$\Delta G^{\circ} AlNbO_4 = \frac{1}{2}(\Delta G^{\circ} Al_2O_3 + \Delta G^{\circ} Nb_2O_5) = -696.kJ \Big/ mol^{-1}O^2 \quad (6)$$

$$\Delta G^{\circ} AlTaO_4 = \frac{1}{2}(\Delta G^{\circ} Al_2O_3 + \Delta G^{\circ} Ta_2O_5) = -725.5kJ \Big/ mol^{-1}O^2 \quad (7)$$

$$\Delta G^{\circ} Ta_8W_9O_{47} = 9\Delta G^{\circ} WO_3 + 4\Delta G^{\circ} Ta_2O_5 = -5400kJ/mol^{-1}O^2 \quad (8)$$

$$\Delta G^{\circ} Nb_8W_9O_{47} = 9\Delta G^{\circ} WO_3 + 4\Delta G^{\circ} Nb_2O_5 = -5270kJ/mol^{-1}O^2 \quad (9)$$

Among the formed simple oxides, WO_3 is easy to sublimate due to its lower sublimation temperature (1000 °C) [57]. For this reason, the concentration of the W atom in OOL is extremely low as compared to that in internal front layer. Simultaneously, the WO_3 will react with Ta_2O_5 to form $Ta_8W_9O_{47}$ due to its high standard free energy of formation. Generally, the formation probability of the complex oxides is not only determined by the standard free energies of formation but also the precursors. $Nb_8W_9O_{47}$ and $Ta_8W_9O_{47}$ has the same crystal structure as well as the concentration of their precursors. However, the absolute value of the standard free energy of formation of $Ta_8W_9O_{47}$ (−5400 ΔG^0 [kJ mole$^{-1}O_2$]) is slightly higher than that of $Nb_8W_9O_{47}$ (−5270 kJ mol$^{-1}O_2$), thus resulting in the formation of a structure involving Nb insertion in $Ta_8W_9O_{47}$ instead of Ta insertion in $Nb_8W_9O_{47}$. Besides, the absolute value of the standard free energy of formation of $Ta_8W_9O_{47}$ (−5400 ΔG^0 [kJ mole$^{-1}O_2$]) is higher than that of $AlTaO_4$ (−725.5 kJ mol$^{-1}O_2$). Thus, the formed Ta_2O_5 will preferentially react with WO_3 to form $Ta_8W_9O_{47}$ instead of $AlTaO_4$. Therefore, there is only little $AlTaO_4$ observed in the IOF. The main oxide phases in the OOL are $AlNbO_4$ and Ti-rich oxides due to the Nb, Al and Ti cations' outward diffusion through the tunnels of the $Ta_8W_9O_{47}$ TTB structure. The W and Ta prefer to form the stable $Ta_8W_9O_{47}$ phase and stay in the IOF. Thus, $AlNbO_4$ and Ti-rich oxides are the dominant phases in the surface of the oxide layer.

5. Conclusion

In this work, a novel oxidation-resistant RHEA WTaNbTiAl was successfully synthesized by the combination of planetary milling and SPS. The RHEA WTaNbTiAl formed a BCC single-phase solid solution with minor oxide and carbide impurities. The Vickers hardness and elastic and shear moduli of WTaNbTiAl RHEA were characterized. In addition, this work provided an insight on the oxidation mechanisms of

this novel RHEA in air at 1000 °C.

The conclusions derived based on the studies conducted in the present work are:

1. The elastic moduli and shear moduli of the WTaNbTiAl are 171 GPa and 65 GPa, respectively. Vickers hardness of the RHEA is 6.7 ± 0.9 GPa.
2. WTaNbTiAl RHEA exhibits promising oxidation resistance with a moderate mass gain of 31.83 mg/cm^2 after 48 h of exposure in air at 1000 °C.
3. The formation of oxide layer (mainly composed of $AlNbO_4$, Ti-riched oxides and $Ta_8W_9O_{47}$) could be linked to oxidation behavior of the WTaNbTiAl alloy. Similar to many oxidation-resistant alloys, the oxidation resistant mechanism of WTaNbTiAl alloy demonstrated that the composite oxide layers are predicted to serve as a reliable diffusion barrier, which effectively prevents the inward diffusion of oxygen and outward diffusion of the cations, thus decreasing its oxidation rate.
4. The mass gain during the oxidation can be modeled with a two-step process - logarithmic and a linear process. A multi-layered structure oxide scale formation process can be used to explain this unique oxidation kinetic. The entire oxide layer exhibits a multi-layered structure after a prolonged exposure to air at 1000 °C, which is composed of an thin OOL ($AlNbO_4$, Ti-rich oxides), an thicker IOF ($Ta_8W_9O_{47}$, Ti-riched oxides), and an POL(tungsten tantalum oxide and Ti-rich oxides). Interestingly, the denser OOL seems to play a key role to prevent the inward diffusion of oxygen, thus improving oxidation resistance of the RHEA WTaNbTiAl. The IOF mainly consisting of $Ta_8W_9O_{47}$ shows a structure of tetragonal tungsten bronzes (TTB), which serves as a path for outward diffusion of the Al, Nb, and Ti cations, which finally form the OOL.

This work gave a delicate nanoscale analysis for the oxidation resistant mechanism of the WTaNbTiAl RHEA, which provides insightful guidance for the oxidation resistant RHEAs design based on the crystal structure of phase in their oxide layers.

CRediT authorship contribution statement

Yonggang Yan: Perform the materials synthesis, Oxidation, Characterization experiment and data analysis, Writing – original draft, Writing – review & editing. **Kade A. McGarrity**: Perform the characterization experiment and data analysis, Writing – review & editing. **Daniel J. Delia**: Perform characterization experiment and data analysis, Writing – review & editing. **Curtis Fekety**: Perform the oxidation and characterization experiment, Writing – review & editing. **Kun Wang:** Conceptualization, Supervision, Conduct the TEM characterization and data analysis, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.corsci.2022.110377.

References

[1] T.M. Pollock, S. Tin, Nickel-based superalloys for advanced turbine engines: chemistry, microstructure and properties, J. Propuls. Power 22 (2006) 361–374.
[2] W. Deng, D. Zhang, H. Wu, Z. Huang, K. Zhou, L. Jiang, Prediction of yield strength in a polycrystalline nickel base superalloy during interrupt cooling, Scr. Mater. 183 (2020) 139–143.
[3] M. Pröbstle, S. Neumeier, P. Feldner, R. Rettig, H.E. Helmer, R.F. Singer, M. Göken, Improved creep strength of nickel-base superalloys by optimized γ/γ′ partitioning behavior of solid solution strengthening elements, Mater. Sci. Eng.: A 676 (2016) 411–420.
[4] D.J. Crudden, A. Mottura, N. Warnken, B. Raeisinia, R.C. Reed, Modelling of the influence of alloy composition on flow stress in high-strength nickel-based superalloys, Acta Mater. 75 (2014) 356–370.
[5] L. Liu, J. Zhang, C. Ai, Nickel-Based Superalloys, in: F.G. Caballero (Ed.), Encyclopedia of Materials: Metals and Alloys, Elsevier, Oxford, 2022, pp. 294–304.
[6] P. Li, W. Jiang, S.-S. Rui, W.-x Yao, H.-j Shi, Q.-n Han, J. Huang, Effect of misorientation on the fatigue life of nickel-base single crystal superalloy DD5 at 980 °C, Int. J. Fatigue 153 (2021), 106479.
[7] H. Pei, M. Li, P. Wang, X. Yao, Z. Wen, Z. Yue, The effect of tensile stress on oxidation behavior of nickel-base single crystal superalloy, Corros. Sci. 191 (2021), 109737.
[8] O.N. Senkov, G.B. Wilks, D.B. Miracle, C.P. Chuang, P.K. Liaw, Refractory high-entropy alloys, Intermetallics 18 (2010) 1758–1765.
[9] O.N. Senkov, D.B. Miracle, K.J. Chaput, J.-P. Couzinie, Development and exploration of refractory high entropy alloys-A review, J. Mater. Res. 33 (2018) 3092–3128.
[10] R. Domagala, R. Ruh, The hafnium-oxygen system, Am. Soc. Met., Trans. Quart. 58 (1965).
[11] T. Schofield, A. Bacon, The constitution of the titanium-oxygen alloys in the range 0-35 weight per cent oxygen, J. Inst. Met. 84 (1955).
[12] E. Gebhardt, H.-D. Seghezzi, W. Dürrschnabel, Untersuchungen in system Zirkonium-Sauerstoff teil 2 untersuchungen zur kinetik der reaktion zwischen Zirkonium und sauerstoff, sowie über die konstitution des systems Zirkonium-Sauerstoff, J. Nucl. Mater. 4 (1961) 255–268.
[13] E.M. Savitskii, Physical Metallurgy of Refractory Metals and Alloys, Springer Science & Business Media,, 2012.
[14] C.M. Liu, H.M. Wang, S.Q. Zhang, H.B. Tang, A.L. Zhang, Microstructure and oxidation behavior of new refractory high entropy alloys, J. Alloy. Compd. 583 (2014) 162–169.
[15] C.-H. Chang, M.S. Titus, J.-W. Yeh, Oxidation Behavior between 700 and 1300 °C of Refractory TiZrNbHfTa High-Entropy Alloys Containing Aluminum, Adv. Eng. Mater. 20 (2018), 1700948.
[16] B. Gorr, F. Mueller, H.-J. Christ, H. Chen, A. Kauffmann, R. Schweiger, D.V. Szabó, M. Heilmaier, Dev. Oxid. Resist. Refract. High. Entropy Alloy. High. Temp. Appl.: Recent Results Dev. Strategy (2018) 647–659.
[17] F. Müller, B. Gorr, H.-J. Christ, H. Chen, A. Kauffmann, M. Heilmaier, Effect of microalloying with silicon on high temperature oxidation resistance of novel refractory high-entropy alloy Ta-Mo-Cr-Ti-Al, Mater. High. Temp. 35 (2017) 168–176.
[18] T.M. Butler, K.J. Chaput, J.R. Dietrich, O.N. Senkov, High temperature oxidation behaviors of equimolar NbTiZrV and NbTiZrCr refractory complex concentrated alloys (RCCAs), J. Alloy. Compd. 729 (2017) 1004–1019.
[19] L.-C. Li, M.-X. Li, M. Liu, B.-Y. Sun, C. Wang, J.-T. Huo, W.-H. Wang, Y.-H. Liu, Enhanced oxidation resistance of MoTaTiCrAl high entropy alloys by removal of Al, Sci. China Mater. 64 (2020) 223–231.
[20] L. Rudolf Kanyane, A. Patricia Popoola, N. Malatji, Development of spark plasma sintered TiAlSiMoW multicomponent alloy: Microstructural evolution, corrosion and oxidation resistance, Results Phys. 12 (2019) 1754–1761.
[21] Y.-K. Kim, Y.-A. Joo, H.S. Kim, K.-A. Lee, High temperature oxidation behavior of Cr-Mn-Fe-Co-Ni high entropy alloy, Intermetallics 98 (2018) 45–53.
[22] Y.Y. Liu, Z. Chen, Y.Z. Chen, J.C. Shi, Z.Y. Wang, S. Wang, F. Liu, Effect of Al content on high temperature oxidation resistance of AlxCoCrCuFeNi high entropy alloys (x=0, 0.5, 1, 1.5, 2), Vacuum 169 (2019), 108837.
[23] T.M. Butler, J.P. Alfano, R.L. Martens, M.L. Weaver, High-Temperature Oxidation Behavior of Al-Co-Cr-Ni-(Fe or Si) Multicomponent High-Entropy Alloys, Jom 67 (2014) 246–259.
[24] B. Gorr, S. Schellert, F. Müller, H.-J. Christ, A. Kauffmann, M. Heilmaier, Current Status of Research on the Oxidation Behavior of Refractory High Entropy Alloys, Adv. Eng. Mater. 23 (2021), 2001047.
[25] O.A. Waseem, U. Auyeskhan, H.M. Lee, H.J. Ryu, A combinatorial approach for the synthesis and analysis of AlxCryMozNbTiZr high-entropy alloys: Oxidation behavior, J. Mater. Res. 33 (2018) 3226–3234.
[26] B. Gorr, F. Müller, M. Azim, H.-J. Christ, T. Müller, H. Chen, A. Kauffmann, M. Heilmaier, High-temperature oxidation behavior of refractory high-entropy alloys: effect of alloy composition, Oxid. Met. 88 (2017) 339–349.
[27] K.-C. Lo, H. Murakami, J.-W. Yeh, A.-C. Yeh, Oxidation behaviour of a novel refractory high entropy alloy at elevated temperatures, Intermetallics 119 (2020), 106711.
[28] T.M. Butler, K.J. Chaput, Native oxidation resistance of Al20Nb30Ta10Ti30Zr10 refractory complex concentrated alloy (RCCA), J. Alloy. Compd. 787 (2019) 606–617.

[29] B. Gorr, F. Mueller, H.J. Christ, T. Mueller, H. Chen, A. Kauffmann, M. Heilmaier, High temperature oxidation behavior of an equimolar refractory metal-based alloy 20Nb 20Mo 20Cr 20Ti 20Al with and without Si addition, J. Alloy. Compd. 688 (2016) 468–477.
[30] O.N. Senkov, S.V. Senkova, D.M. Dimiduk, C. Woodward, D.B. Miracle, Oxidation behavior of a refractory NbCrMo0.5Ta0.5TiZr alloy, J. Mater. Sci. 47 (2012) 6522–6534.
[31] K.C. Lo, Y.J. Chang, H. Murakami, J.W. Yeh, A.C. Yeh, An oxidation resistant refractory high entropy alloy protected by CrTaO4-based oxide, Sci. Rep. 9 (2019) 7266.
[32] B. Gorr, F. Müller, S. Schellert, H.-J. Christ, H. Chen, A. Kauffmann, M. Heilmaier, A new strategy to intrinsically protect refractory metal based alloys at ultra high temperatures, Corros. Sci. 166 (2020), 108475.
[33] M. Allouard, Y. Bienvenu, L. Nazé, C.B. Brancho-Troconis, Development of oxidation resistant high temperature NbTiAl alloys and intermetallics, Le. J. De. Phys. IV 03 (1993) C9-419–C419-428.
[34] P. Zhang, Y. Li, Z. Chen, J. Zhang, B. Shen, Oxidation response of a vacuum arc melted NbZrTiCrAl refractory high entropy alloy at 800–1200 °C, Vacuum 162 (2019) 20–27.
[35] S. Schellert, B. Gorr, H.J. Christ, C. Pritzel, S. Laube, A. Kauffmann, M. Heilmaier, The Effect of Al on the Formation of a CrTaO4 Layer in Refractory High Entropy Alloys Ta-Mo-Cr-Ti-xAl, Oxid. Met. 96 (2021) 333–345.
[36] Y. Yan, D. Lu, K. Wang, Accelerated discovery of single-phase refractory high entropy alloys assisted by machine learning, Comput. Mater. Sci. 199 (2021), 110723.
[37] O.A. Waseem, H.J. Ryu, Powder Metallurgy Processing of a WxTaTiVCr High-Entropy Alloy and Its Derivative Alloys for Fusion Material Applications, Sci. Rep. 7 (2017) 1926.
[38] M. Krzyzanowski, J.H. Beynon, D.C.J. Farrugia, (2010).
[39] H. Shi, C. Tang, A.D. Jianu, R. Fetzer, A. Weisenburger, M. Steinbrueck, M. Grosse, R. Stieglitz, G. Müller, Oxidation behavior and microstructure evolution of alumina-forming austenitic & high entropy alloys in steam environment at 1200/°C, Corros. Sci. 170 (2020), 108654.
[40] N. Birks, G.H. Meier, F.S. Pettit, Introduction to the High Temperature Oxidation of Metals, Cambridge University Press,, 2006.
[41] A. Ostovari Moghaddam, M. Sudarikov, N. Shaburova, D. Zherebtsov, V. Zhivulin, I. Ashurali Solizoda, A. Starikov, S. Veselkov, O. Samoilova, E. Trofimov, High temperature oxidation resistance of W-containing high entropy alloys, J. Alloy. Compd. 897 (2022), 162733.
[42] M. Ghadyani, C. Utton, P. Tsakiropoulos, Microstructures and Isothermal Oxidation of the Alumina Scale Forming Nb1.7Si2.4Ti2.4Al(3)Hf0.5 and Nb1.3Si2.4Ti2.4Al3.5Hf0.4 Alloys, Mater. (Basel) 12 (2019).
[43] R. Wang, Y. Tang, S. Li, Y. Ai, Y. Li, B. Xiao, La Zhu, X. Liu, S. Bai, Effect of lattice distortion on the diffusion behavior of high-entropy alloys, J. Alloy. Compd. 825 (2020), 154099.
[44] R. Zhang, J. Meng, J. Han, K. Tulugan, R. Zhang, Oxidation resistance properties of refractory high-entropy alloys with varied AlxCrTiMo content, J. Mater. Sci. 56 (2020) 3551–3561.
[45] S. Lu, X. Li, X. Liang, W. Yang, J. Chen, Effect of V and Ti on the Oxidation Resistance of WMoTaNb Refractory High-Entropy Alloy at High Temperatures, Metals 12 (2021) 41.
[46] S. Lu, X. Li, X. Liang, W. Shao, W. Yang, J. Chen, Effect of Al content on the oxidation behavior of refractory high-entropy alloy AlMo0.5NbTa0.5TiZr at elevated temperatures, Int. J. Refract. Met. Hard Mater. 105 (2022), 105812.
[47] Y. Du, D. Ding, L. Lai, S. Xiao, N. Guo, B. Song, S. Guo, Effect of Y on the high-temperature oxidation behavior of CrMoTaTi refractory high entropy alloy, Int. J. Refract. Met. Hard Mater. 103 (2022), 105755.
[48] D. Ouyang, Z.-j Chen, H.-b Yu, K.C. Chan, L. Liu, Oxidation behavior of the Ti38V15Nb23Hf24 refractory high-entropy alloy at elevated temperatures, Corros. Sci. 198 (2022), 110153.
[49] J.R. DiStefano, B.A. Pint, J.H. DeVan, Oxidation of refractory metals in air and low pressure oxygen gas, Int. J. Refract. Met. Hard Mater. 18 (2000) 237–243.
[50] F. Müller, B. Gorr, H.-J. Christ, J. Müller, B. Butz, H. Chen, A. Kauffmann, M. Heilmaier, On the oxidation mechanism of refractory high entropy alloys, Corros. Sci. 159 (2019), 108161.
[51] J. Gild, Y. Zhang, T. Harrington, S. Jiang, T. Hu, M.C. Quinn, W.M. Mellor, N. Zhou, K. Vecchio, J. Luo, High-entropy metal diborides: a new class of high-entropy materials and a new type of ultrahigh temperature ceramics, Sci. Rep. 6 (2016) 37946.
[52] F. Krumeich, Intergrowth of niobium tungsten oxides of the tetragonal tungsten bronze type, Z. für Naturforsch. B 75 (2020) 913–919.
[53] S.M. Montemayor, A.A. Mendez, A. Martínez-de la Cruz, A.F. Fuentes, L.M. Torres-Martínez, Lithium insertion in two tetragonal tungsten bronze type phases, M8W9O47 (M=Nb and Ta), J. Mater. Chem. 8 (1998) 2777–2781.
[54] J. Jayaraj, P. Thirathipviwat, J. Han, A. Gebert, Microstructure, mechanical and thermal oxidation behavior of AlNbTiZr high entropy alloy, Intermetallics 100 (2018) 9–19.
[55] E. Osei-Agyemang, G. Balasubramanian, Surface oxidation mechanism of a refractory high-entropy alloy, npj Mater. Degrad. 3 (2019).
[56] H. Jehn, E. Olzi, High temperature solid-solubility limit and phase studies in the system tantalum-oxygen, J. Less Common Met. 27 (1972) 297–309.
[57] B. Gorr, M. Azim, H.J. Christ, T. Mueller, D. Schliephake, M. Heilmaier, Phase equilibria, microstructure, and high temperature oxidation resistance of novel refractory high-entropy alloys, J. Alloy. Compd. 624 (2015) 270–278.
[58] S. Sheikh, L. Gan, T.-K. Tsao, H. Murakami, S. Shafeie, S. Guo, Aluminizing for enhanced oxidation resistance of ductile refractory high-entropy alloys, Intermetallics 103 (2018) 40–51.
[59] Y.-k Cao, Y. Liu, B. Liu, W.-d Zhang, J.-w Wang, M. Du, Effects of Al and Mo on high temperature oxidation behavior of refractory high entropy alloys, Trans. Nonferrous Met. Soc. China 29 (2019) 1476–1483.
[60] N. Yurchenko, E. Panina, S. Zherebtsov, G. Salishchev, N. Stepanov, Oxidation Behavior of Refractory AlNbTiVZr0.25 High-Entropy Alloy, Mater. (Basel) 11 (2018).
[61] S.K. Varma, F. Sanchez, S. Moncayo, C.V. Ramana, Static and cyclic oxidation of Nb-Cr-V-W-Ta high entropy alloy in air from 600 to 1400 °C, J. Mater. Sci. Technol. 38 (2020) 189–196.
[62] Y. Xiao, W. Kuang, Y. Xu, L. Wu, W. Gong, J. Qian, Q. Zhang, Y. He, Microstructure and oxidation behavior of the CrMoNbTaV high-entropy alloy, J. Mater. Res. 34 (2018) 301–308.
[63] F. Müller, B. Gorr, H.-J. Christ, H. Chen, A. Kauffmann, M. Heilmaier, Effect of Y Additions on the Oxidation Behaviour of Novel Refractory High-Entropy Alloy NbMoCrTiAl at 1000 °C in Air, Oxid. Met. 94 (2020) 147–163.
[64] P.Y. Hou, Oxid. Met. Alloy. (2010) 195–239.
[65] O.A. Waseem, H.J. Ryu, Combinatorial synthesis and analysis of AlxTayVz-Cr20Mo20Nb20Ti20Zr10 and Al10CrMoxNbTiZr10 refractory high-entropy alloys: Oxidation behavior, J. Alloy. Compd. 828 (2020), 154427.